



PHOENIX, ARIZONA

CST-239
Milestone 2

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Screencast link:

<https://www.loom.com/share/90f04f474cc64a7b9e301c9ff1b4d28b>

UML:

StoreFront Class Diagram with Inventory and Product Hierarchy

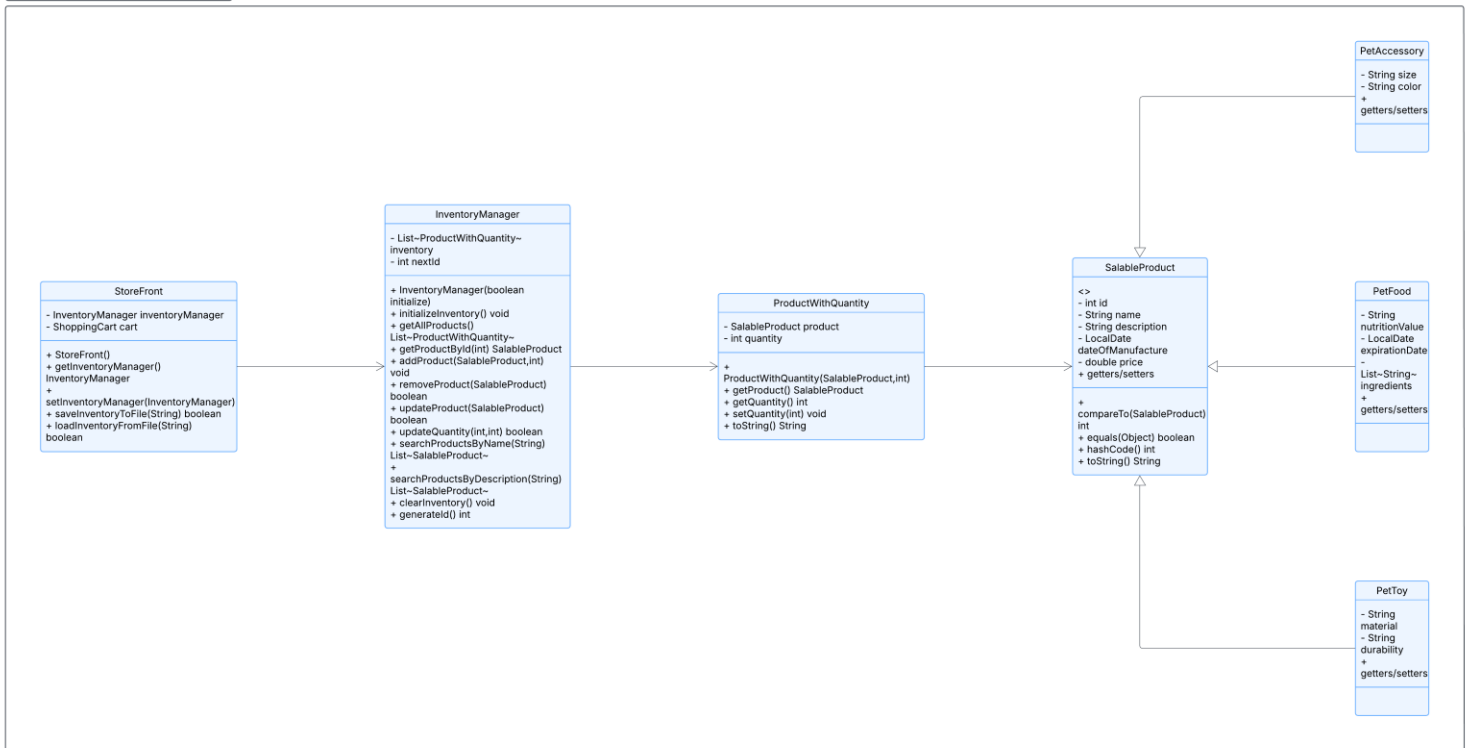


Figure 1 UML Diagram

Flowchart:

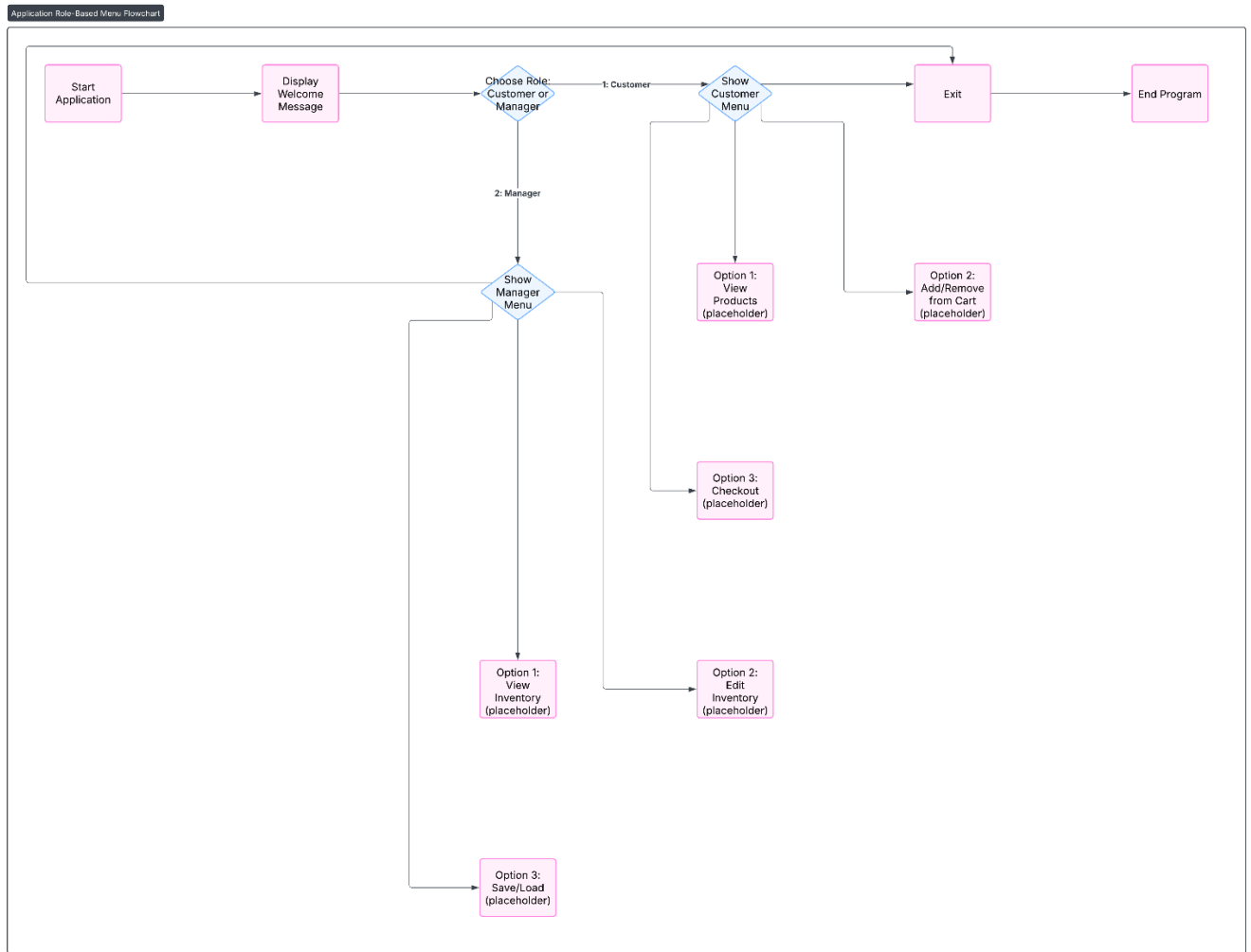


Figure 2 Program Flowchart

Output Screenshots:

```
Select option: 4

Choose product type to add:
1) PetFood
2) PetToy
3) PetAccessory
Enter 1-3: 1
Name: Gravy
Description: Gravy flavoured cans
Date of Manufacture (format: yyyy-MM-dd): 2025-05-10
Price: 12
Nutrition Value: 200 calorie
Expiration Date (format: yyyy-MM-dd): 2026-05-10
Add ingredient (blank to finish):
Starting quantity: 10
Added: Product{id=5, name='Gravy', price=12.00}, Qty=10

--- Store Manager Menu ---
1) View Inventory
2) Search by Name
3) Search by Description
4) Add Product
5) Edit Product
6) Remove Product
7) Update Quantity
8) Clear Inventory
9) Save Inventory (stub)
10) Load Inventory (stub)
0) Back to Main Menu
Select option: 1
Product{id=1, name='Crunchy Kibble', price=24.99}, Qty=15
Product{id=2, name='Squeaky Ball', price=8.49}, Qty=30
Product{id=3, name='Comfort Collar', price=12.99}, Qty=20
Product{id=4, name='Gourmet Wet Food', price=3.49}, Qty=40
Product{id=5, name='Gravy', price=12.00}, Qty=10
```

Figure 3 Adding a product

```

4) Add Product
5) Edit Product
6) Remove Product
7) Update Quantity
8) Clear Inventory
9) Save Inventory (stub)
10) Load Inventory (stub)
0) Back to Main Menu
Select option: 7
Product{id=1, name='Crunchy Kibble', price=24.99}, Qty=15
Product{id=2, name='Squeaky Ball', price=8.49}, Qty=30
Product{id=3, name='Comfort Collar', price=12.99}, Qty=20
Product{id=4, name='Gourmet Wet Food', price=3.49}, Qty=40
Product{id=5, name='Gravy', price=12.00}, Qty=10
Enter product ID to change quantity: 1
New quantity (>=0): 20
Quantity updated.

--- Store Manager Menu ---
1) View Inventory
2) Search by Name
3) Search by Description
4) Add Product
5) Edit Product
6) Remove Product
7) Update Quantity
8) Clear Inventory
9) Save Inventory (stub)
10) Load Inventory (stub)
0) Back to Main Menu
Select option: 1
Product{id=1, name='Crunchy Kibble', price=24.99}, Qty=20
Product{id=2, name='Squeaky Ball', price=8.49}, Qty=30
Product{id=3, name='Comfort Collar', price=12.99}, Qty=20
Product{id=4, name='Gourmet Wet Food', price=3.49}, Qty=40
Product{id=5, name='Gravy', price=12.00}, Qty=10

```

Figure 4 Update Quantity

```

6) Remove Product
7) Update Quantity
8) Clear Inventory
9) Save Inventory (stub)
10) Load Inventory (stub)
0) Back to Main Menu
Select option: 6
Product{id=1, name='Crunchy Kibble', price=24.99}, Qty=20
Product{id=2, name='Squeaky Ball', price=8.49}, Qty=30
Product{id=3, name='Comfort Collar', price=12.99}, Qty=20
Product{id=4, name='Gourmet Wet Food', price=3.49}, Qty=40
Product{id=5, name='Gravy', price=12.00}, Qty=10
Enter product ID to remove: 5
Removed.

--- Store Manager Menu ---
1) View Inventory
2) Search by Name
3) Search by Description
4) Add Product
5) Edit Product
6) Remove Product
7) Update Quantity
8) Clear Inventory
9) Save Inventory (stub)
10) Load Inventory (stub)
0) Back to Main Menu
Select option: 1
Product{id=1, name='Crunchy Kibble', price=24.99}, Qty=20
Product{id=2, name='Squeaky Ball', price=8.49}, Qty=30
Product{id=3, name='Comfort Collar', price=12.99}, Qty=20
Product{id=4, name='Gourmet Wet Food', price=3.49}, Qty=40

```

Figure 5 Remove Product

```
--- Store Manager Menu ---
1) View Inventory
2) Search by Name
3) Search by Description
4) Add Product
5) Edit Product
6) Remove Product
7) Update Quantity
8) Clear Inventory
9) Save Inventory (stub)
10) Load Inventory (stub)
0) Back to Main Menu
Select option: 2
Enter name search term: ball
Product{id=2, name='Squeaky Ball', price=8.49}, Qty=30
```

Figure 6 Search for product